

Krishna Baheti

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EDUCATION

Sardar Vallabhbhai National Institute of Technology, Surat

Surat, India

B.Tech in Artificial Intelligence (CGPA: 9.08/10)

2024 – 2028

CBSE Class XII

Jalgaon, India

District Topper – 96.20%

2023 – 2024

RELEVANT COURSEWORK

Data Structures and Algorithms, Operating Systems, Computer Organisation and Architecture, Object Oriented Programming, Computer Networks, Database Management Systems.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, SQL

Systems & Backend: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Redis, BullMQ, Concurrency, Distributed Systems

Databases & Data: PostgreSQL, MongoDB, Redis, pgvector

AI/ML: Retrieval-Augmented Generation (RAG), Vector Search, Embeddings, XGBoost

PROJECTS

CodeSync – Real-Time Collaborative Code Editor | Next.js Node.js, WebSockets, Yjs CRDT, Redis, PostgreSQL
[Code](#) | [Architecture](#) 2026

- Implemented conflict-free collaborative editing using Yjs CRDT, enabling concurrent edits, offline synchronization, and automatic state reconciliation across clients.
- Designed a distributed WebSocket architecture using Redis Pub/Sub and binary message serialization to synchronize collaborative editing across multiple backend instances.
- Implemented persistent document storage using PostgreSQL snapshots and event logs with role-based access control for secure multi-user collaboration and version history.
- Built collaborative coding features including interview rooms, WebRTC audio, anonymous demo sessions, and a dashboard for monitoring collaborative editing events.

Smart Ride Pooling Dispatch Engine | Next.js, Express.js, Redis, BullMQ, PostgreSQL

[Code](#) | [Architecture](#)

2026

- Modeled real-time ride dispatch as a Vehicle Routing Problem (VRP) with rider detour and vehicle capacity constraints.
- Implemented ride matching using greedy compatibility scoring followed by 2-opt local search to improve route assignments under routing constraints.
- Designed an event-driven backend using Redis queues, BullMQ workers, Socket.IO, and PostgreSQL to process ride requests asynchronously and synchronize application state.
- Integrated an XGBoost-based demand prediction module to estimate ride demand and support driver repositioning.

ChatPDF – AI-Powered Document Assistant | Next.js, FastAPI, Gemini, Supabase, pgvector

[Code](#) | [Architecture](#)

2026

- Built an end-to-end Retrieval-Augmented Generation (RAG) system using Gemini embeddings, pgvector, and Supabase for conversational document retrieval.
- Implemented semantic vector retrieval, conversation-aware context management, and embedding pipelines for document search.
- Engineered citation-aware PDF navigation by mapping generated responses to exact document regions using page-level bounding boxes.
- Implemented retry mechanisms, rate-limit handling, JWT-based authentication, and user-level document isolation for secure document access.

ACHIEVEMENTS

- LeetCode Knight (1900+ Rating) with 400+ algorithmic problems solved across advanced Data structures & Algorithms.
- Codeforces Expert (1600+ Rating), demonstrating strong competitive programming and problem-solving skills.
- Selected for Amazon ML Summer School (2026), among the top 3,000 participants from 130,000+ nationwide registrations (Top 2.3%).
- Second Runner-Up, Adani Powermind Hackathon for developing **ChatPDF**.

POSITIONS OF RESPONSIBILITY

- Junior Developer, Google Developer Groups (NIT Surat)
- Representative, ACM Student Chapter (NIT Surat)